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$$\tan(\text{Bgsin } x)$$

Stel:

$$t = \text{Bgsin } x, \quad t \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

dan geldt:

$$\begin{aligned} \tan t &= \frac{\sin t}{\cos t} \\ &= \frac{\sin(\text{Bgsin } x)}{\cos(\text{Bgsin } x)} \\ &= \frac{\sin(\text{Bgsin } x)}{\sqrt{1 - \sin^2(\text{Bgsin } x)}} \\ &= \frac{x}{\sqrt{1 - x^2}} \text{ met } x \in [-1, 1] \end{aligned}$$

References